ACCENTURE INNOVATION CHALLENGE 2026 · ROUND 2 · PROBLEM TRACK 4

Team : Catalyst_IITB_DigitalTwin

DigitalTwin.ai

A Predictive-Prescriptive Digital Twin for Vehicle Assembly Line Operations

Detailed Business Proposal — Prepared for Round 2 Evaluation

TEAM INTRODUCTION

Meet the Team Behind DigitalTwin.ai

Team : Catalyst_IITB_DigitalTwin | Accenture Innovation Challenge 2026, Problem Track 4

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Overall Outline

This proposal presents DigitalTwin.ai, a predictive-prescriptive digital twin developed for vehicle assembly line operations under Problem Track 4 of the Accenture Innovation Challenge 2026. The solution addresses a persistent and costly gap in manufacturing operations, i.e. the inability to detect bottlenecks and defects before they propagate downstream and surface as scrap, rework, or unplanned line stoppage.

Using a graph-based forecasting model, DigitalTwin.ai predicts station-level operating conditions five to thirty minutes in advance, identifies the station at which an emerging anomaly most likely originated, and recommends a corrective action to the appropriate operating role, based on a premade rule book. The prototype has been validated on an eighteen-station simulated assembly line incorporating a realistic distribution of sensor coverage, and demonstrates a measurable improvement in forecast accuracy when inter-station dependencies are modelled explicitly.

This document sets out the business rationale, technical approach, expected impact, phased implementation plan, and associated risks, and concludes with a request for a ninety-day pilot engagement on a single production line to validate the solution under real operating conditions.

23%

Reduction in downstream forecast error when station interdependencies are modelled

18

Station simulated line validated in the current prototype; the approach extends to 30–50

The Cost of Detecting Problems Too Late

Modern vehicle assembly lines operate as sequences of thirty to fifty interdependent stations spanning body construction, paint, and final assembly. A deviation at any single station a gradual increase in cycle time, a drift in process parameters, or an intermittent quality defect rarely remains isolated. It propagates downstream and frequently surfaces only at a later inspection point, by which time a substantial number of units may already carry the same undetected condition.

Existing monitoring infrastructure in most plants is reactive by design, alerts and quality flags are generated after a fault has already occurred, at which point the associated cost, in scrap, rework, missed shipments, or unplanned stoppage, has already been incurred.

A further complicating factor is that most assembly lines are **not uniformly instrumented**. They represent a layered accumulation of equipment installed across different periods, resulting in **stations with modern sensor arrays alongside stations with only partial or indirect signal availability**, and others that rely entirely on manual checklists. A solution intended for real-world deployment must operate credibly across this uneven instrumentation landscape, rather than assuming an idealised, fully sensored line.

$20K–$50K

Lost per minute of unplanned downtime on a high-volume automotive assembly line

Source: manufacturing-cost benchmarks (Aberdeen Research, Siemens, industry press), 2025–26

Instrumentation reality: On a typical assembly line, roughly one in four stations provides only partial or manual signal coverage, and physical upgrades are constrained to a small number of scheduled maintenance windows per year.

PROPOSED SOLUTION

A Predictive Layer Alongside Existing Plant Systems

DigitalTwin.ai is proposed as a continuous, predictive layer positioned alongside existing plant systems rather than as a replacement for them. Its function is to observe the operating state of every station on the line, forecast near-term deviations before they manifest as measurable defects or slowdowns, and communicate both the nature and probable origin of an emerging issue to plant personnel, together with a recommended corrective action. ***This structure allows the system to function meaningfully even where sensor coverage is incomplete.***

1

Acquire

Collect operating signals from every station, regardless of instrumentation level.

→

2

Estimate

Fill gaps at under-instrumented stations using estimated sensor values.

→

3

Model

Represent the line as a graph so that dependencies between stations are captured explicitly.

→

4

Forecast

Predict station-level operating conditions across short and long time horizons.

→

5

Recommend

Generate a prioritised, explainable recommendation for the operating team.

Core Forecasting Mechanism

The forecasting core combines two complementary modelling techniques. A station-level encoder, implemented as a Long Short-Term Memory (LSTM) network, learns the temporal operating pattern of each individual station from its own historical signal. This is followed by a Graph Convolutional Network (GCN) layer, which propagates the learned state of each station to its immediate downstream neighbour, reflecting the physical sequence of the assembly line. This combined design allows the model to anticipate a downstream disturbance before it physically arrives, rather than waiting for the disturbance to appear in that station's own sensor history.

Anomalies are identified through residual-based thresholding against each station's expected operating range, and each flagged anomaly is attributed to its most probable originating station using timing and network-propagation analysis providing operators with an explanation, not an undifferentiated alert.

23% lower

Forecast error two stations downstream of an induced bottleneck, versus a station-level model with no cross-station awareness

Forecast Horizons

Short horizon: 5-minute forecast, for immediate operator response
Long horizon: 30-minute forecast, for supervisory and planning use

Estimating Missing Sensor Data at Under-Instrumented Stations

A meaningful proportion of stations on any real assembly line will not be equipped with the direct sensors required to measure the condition of interest. In the validated prototype, approximately seventeen percent of stations provide only indirect, or proxy, signals, and a further eleven percent are monitored through manual checklists alone.

To address this **without requiring immediate capital investment** in additional instrumentation, DigitalTwin.ai incorporates a dedicated **soft-sensor estimation layer**. This component is trained on stations where both true sensor readings, such as torque, vibration, and temperature, and indirect signals are simultaneously available. Using a gradient-boosting regression model, it learns the statistical relationship between indirect signals including motor current draw, control-loop setpoint error, and station cycle time and the true sensor values with which they correlate.

Once trained, this model is applied to stations that lack direct instrumentation, producing estimated sensor values in real time from the indirect signals already present at that station. These estimated values are then treated as first-class inputs to the downstream forecasting model, on the same basis as directly measured readings.

Indirect Signals Used

- Motor current draw
- Control-loop setpoint error
- Station cycle time

Validation approach: true-sensor stations are withheld during training and used to confirm that soft-sensor estimates track ground truth within an acceptable margin, before the model is trusted at fully unsensored stations.

Data Capture and Continuous Improvement Framework

The predictive accuracy of a data-driven system of this nature is expected to improve as it is exposed to a larger and more varied set of operating conditions than any single production line can provide in isolation. This proposal therefore includes an optional, governed data collection framework intended to support the long-term robustness of the platform across deployments.

1 Consent-Based Participation

Any cross-site data aggregation is strictly optional and remains within the client's control, with the ability to withdraw consent and have contributed data excluded from future model training at any time.

2 Defined & Limited Scope

Data collected is scoped under an explicit, written data use agreement, limited to operational parameters sensor and proxy signal values, cycle timing, and confirmed quality outcomes and excludes any personally identifiable information.

3 Anonymisation at Source

All data used for centralised model improvement is anonymised at the point of collection, with plant and unit identifiers removed prior to storage.

4 Independent Base Deployment

The base solution is fully functional using only a client's own operating data, retained within their own environment, without any requirement to participate in cross-site data sharing.

STAKEHOLDER VALUE

A Single Model, Serving Three Operating Perspectives

The underlying forecasting model is shared across the organisation, but its output is presented differently to each role, in keeping with the decision that role is responsible for making.

FS

Floor Supervisor

REAL-TIME OPERATING VIEW

Receives live alerts at the moment a station begins to drift from its expected pattern, accompanied by a plain-language explanation and a recommended immediate action, formatted for review within seconds during an active shift.

PM

Plant Manager

PLANNING VIEW

Reviews trends across stations and shifts, tracks recurring problem areas, and schedules maintenance windows in line with patterns the twin has been forecasting over the preceding period.

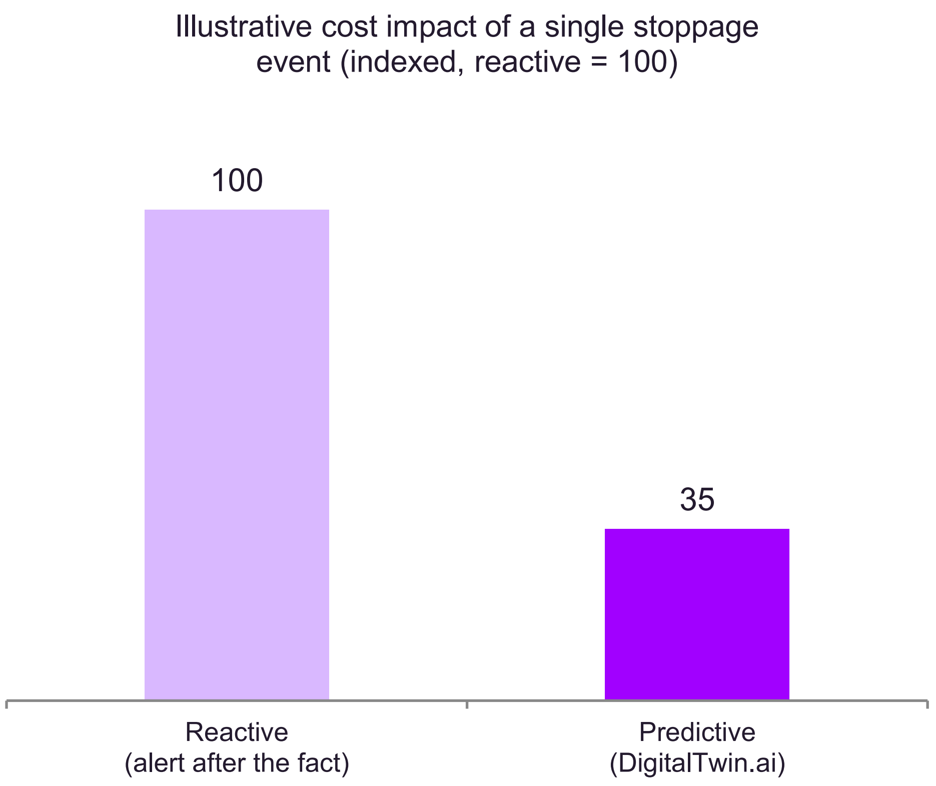
LD

Leadership

STRATEGIC INVESTMENT VIEW

Receives a consolidated business case summarising cost avoided, uptime gained, and return on investment, to support decisions on extending the platform to additional lines and plants.
Later, NLP model can be used to make this more dynamic and user centric.

The Financial Rationale for Early Detection



Reduced downtime exposure

Industry benchmarks indicate that reducing unplanned downtime by as little as ten minutes a day on a high-value line can be worth in excess of two hundred thousand dollars a day; early warning is the mechanism by which those minutes are recovered.

Reduced downstream scrap and rework

Identifying a defect at its originating station, rather than at final inspection, prevents the same condition from repeating across the dozens of units that would otherwise pass through the affected station before detection.

Higher operator confidence and adoption

Each alert is accompanied by a confidence score and a stated reason, allowing floor teams to evaluate why a station was flagged rather than treating the system as an unexplained black box, which supports sustained adoption.

IMPLEMENTATION ROADMAP

A Phased Path to Enterprise Deployment

0 1 2 3

CURRENT STATE

Prototype

A simulator, forecasting model, and three-tier dashboard have been built and validated end to end on realistic simulated production data.

MONTHS 0–3

Initial Deployment

Connect the platform to one production line's live sensor feeds. Validate forecasts against real outcomes and calibrate alert thresholds jointly with floor staff.

MONTHS 3–6

Plant-Wide Scale-Up

Extend coverage to every station and line within the plant, and incorporate floor-team overrides as a structured feedback loop for ongoing refinement.

MONTHS 6–12

Multi-Site Expansion

Extend deployment to additional plants with differing layouts, and evolve the fixed rule-based action engine toward a learning-based recommendation system.

Principal Risks and Corresponding Mitigations

RISK	MITIGATION
False positives undermine operator trust	Every alert carries an explicit confidence score, thresholds remain tunable by plant staff, and no automated action is taken without human review; the system is deliberately biased toward escalation rather than silent downgrade under uncertainty.
Integration risk with live production systems	The platform integrates on a read-only basis through OT gateways, without modification to programmable logic controllers or existing control logic; any physical instrumentation change is scheduled within existing maintenance windows only.
Variation in layout across plants and lines	The underlying model is graph-based and configuration-driven; new stations, lines, or plants are incorporated as additional graph nodes rather than requiring the model to be rebuilt for each site.
Forecast accuracy degrading over time	Forecast outcomes are continuously validated against actual results, with retraining can be triggered automatically once observed accuracy falls below a defined threshold.

Recommendation and Next Steps

DigitalTwin.ai has been demonstrated, on a working prototype, to predict station-level conditions in advance, to explain the origin of an emerging issue, and to recommend a corrective response. The design accounts explicitly for uneven sensor coverage and for the operational constraints of live production environments.

The recommended next step is a controlled pilot to validate these results under real operating conditions, prior to any decision on wider deployment.

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github.com/ApratimAnand336/Digitwin